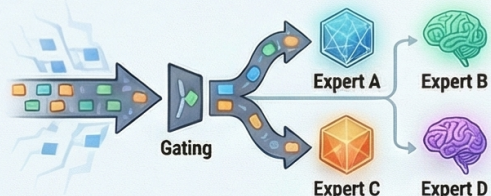


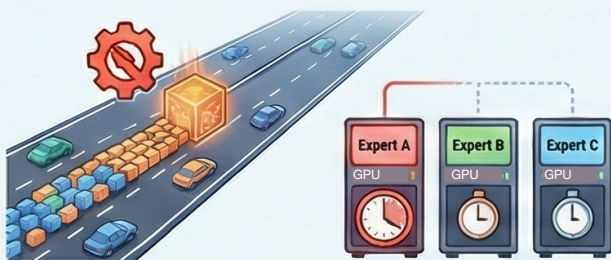
The Problem: The MoE Expert Traffic Jam

MoE Models Use "Experts" to Work Smarter



MoE architecture replaces standard network layers with a set of specialized "experts" networks. A gating network routes incoming data to a small subset of these experts for processing, reducing computational load

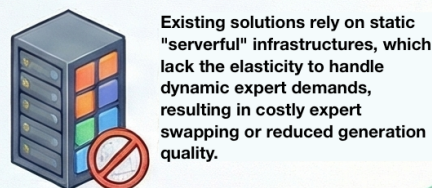
Expert Load Imbalance Creates a Bottleneck



Overloaded Experts Become "Stragglers"

GPUs hosting popular experts take much longer to finish their work, forcing other GPUs with lightly loaded experts to wait, which severely increases inference latency.

Traditional Solutions are Inefficient

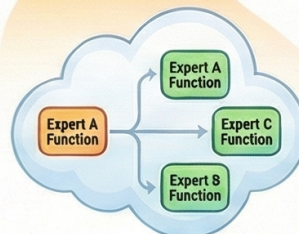


Step 1: Proactive Load Prediction

A lightweight, layer aware predictor accurately estimates the upcoming load distribution for each expert in advance, identifying potential stragglers before they occur.

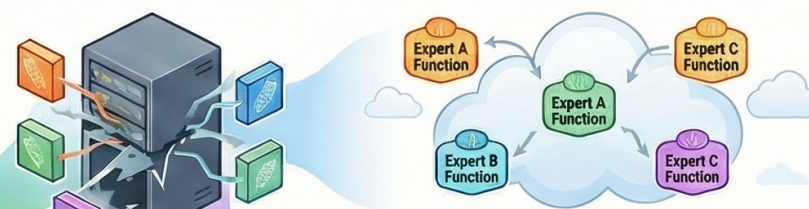
Step 2: Dynamic Expert Scaling

Based on predictions, the Expert Scaler automatically adds or removes replicas of experts to perfectly match the anticipated workload and eliminate bottlenecks.



The Solution: The ServerlessMoE Framework

Decoupling Experts with Serverless Functions

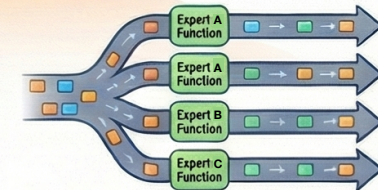


ServerlessMoE packages individual experts as independent, stateless functions that can be scaled and deployed on demand, enabling true elasticity.



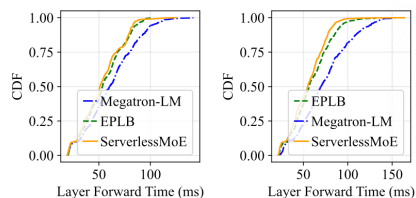
Step 3: Optimized Expert Placement

The Expert Placer strategically assigns expert replicas to GPUs to maximize reuse (warm-starts), balance the computational load across all GPUs, and minimize communication overhead.



Step 4: Balanced Expert Serving

The workload for each expert is evenly distributed across its replicas, allowing for parallel processing that eliminates stragglers and accelerates inference.



(a) Mixtral-8x7B.

(b) Phi-3.5-MoE.

Figure 8: MoE layer forward time of three approaches.

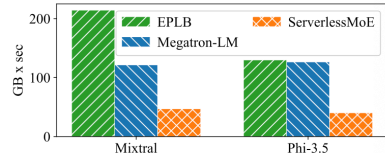


Figure 9: Total inference cost of three approaches.

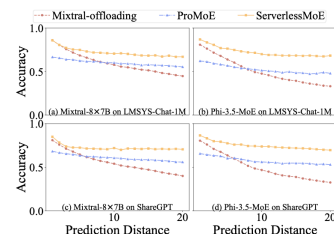


Figure 10: Expert load prediction accuracy for three prediction methods at different prediction distances.

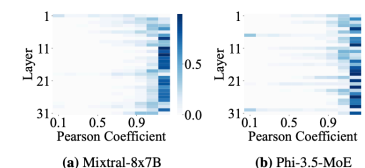
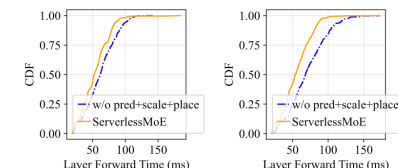


Figure 11: Correlations between predicted and actual expert load distributions across layers of two models. Heavier color means more correlation results fall in the slot.



(a) Mixtral-8x7B.

(b) Phi-3.5-MoE.

Figure 12: MoE layer forward time of ablation study.